Using Open Source Intelligence Techniques to evaluate Physics Calculations

Physics students and Nathan T. Moore

Physics and General Engineering, Winona State University, Winona, MN 55987 USA

E-mail: `nmoore@winona.edu`

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Abstract. Checking the answer at the end of a calculation is an oft neglected part of solving physics problems, and getting students to engage in this behavior can be difficult when they are already intellectually tired. The paper briefly describes a graphing, linearization, and control of variables activity related to pendulums, couched as "swingset" design. After a reliable $swingtime \sim \sqrt{length}$ model has been established, students apply the model to a youtube video to predict the height of a bridge. Then, to reinforce the ideas of multiple representations, and multiple lines of evidence within an argument, students are asked to use Google Maps and other online resources to estimate the height of the bridge. Looking for overlap between online estimations and physical models from the lab is powerful. At the end of the exercise, students have had a brief introduction to "Open Source Intelligence" techniques that are commonly used by news and intelligence organizations.

1. Swingset Exercise

Early in a Physical Science or Introductory Physics-Mechanics class, we introduce and practice control of variables, graphing, and linearization concepts by asking students to think about designing a new swingset for an elementary school playground. An exciting swingset takes a long time to swing back and forth and we call this "swingtime" the output variable we want to be able to predict. Formally, a swingset is typically modeled as a pendulum with period $T \sim \sqrt{L}$ [1, 2] but as this activity usually occurs in the first few weeks of the class, we avoid any mention of the technical story that for small oscillations a pendulum's period is $T \simeq 2\pi\sqrt{L/g}$.

When asked about possible input variables on which swingtime might depend, students will often talk about mass, weight, initial height and angle, whether you start from rest or with an initial push, and length. A video introduction to Estonia sport of Kiiking [3] can be a great anchoring event for this disucssion. After brief investigations, the students typically settle on initial angle, mass, and length as the most important variables needed to predict swingtime. With further data collection, the class can narrow down the length of the swing's cable as the most important variable.

Figure 1. In North America, a "swingset" or "swing" is a mass-rope pendulum system that people enjoy using. Swings have had an astonishingly long history of recreational use throughout many human cultures [5].

Data from a past class is given in figure 2. Thus far, all of the students' measurements have been collected with metal washers and string, with none of the swingsets larger than about $100cm$ in length. The initial data in figure 2 looks fairly linear, and when asked to predict the swingtime for a $200cm$ length, students typically either draw a line or use a linear math model, also shown in figure 2.

This linear model fails at longer swingset lengths. Resolution comes via two pieces of data. First, if a swingset has "almost no length" students will generally volunteer that it takes "almost no time" to swing back and forth. This means that the $length = 0$, $swingtime = 0$ intercept is physically real and should be included in fitting efforts.

Second, when results from swingset lengths of $200cm$, $300cm$ and $400cm$ are included in the graph, students can be coached to recollect that the data reminds them of a parabola that has been rotated to open towards the positive x-axis of the graph. In Pre-Calculus or Algebra 2 classes this mathematical form is often written as a $y^2 = x$ or $y = \sqrt{x}$ relationship.

To test this "curvy" relationship between swingtime and length, we ask the students to create the graph shown in figure 3, in which the data has been "linearized" by plotting $\sqrt{L}$ instead of L. In our experience, many students have not seen a non-linear proportionality "in the wild" before, and this can be a challenging concept to discuss – particularly when thinking about the units involved in linearized trendlines!

A fun final test of the linearized model for swingtime is to use a $10m$ surveyor's tape

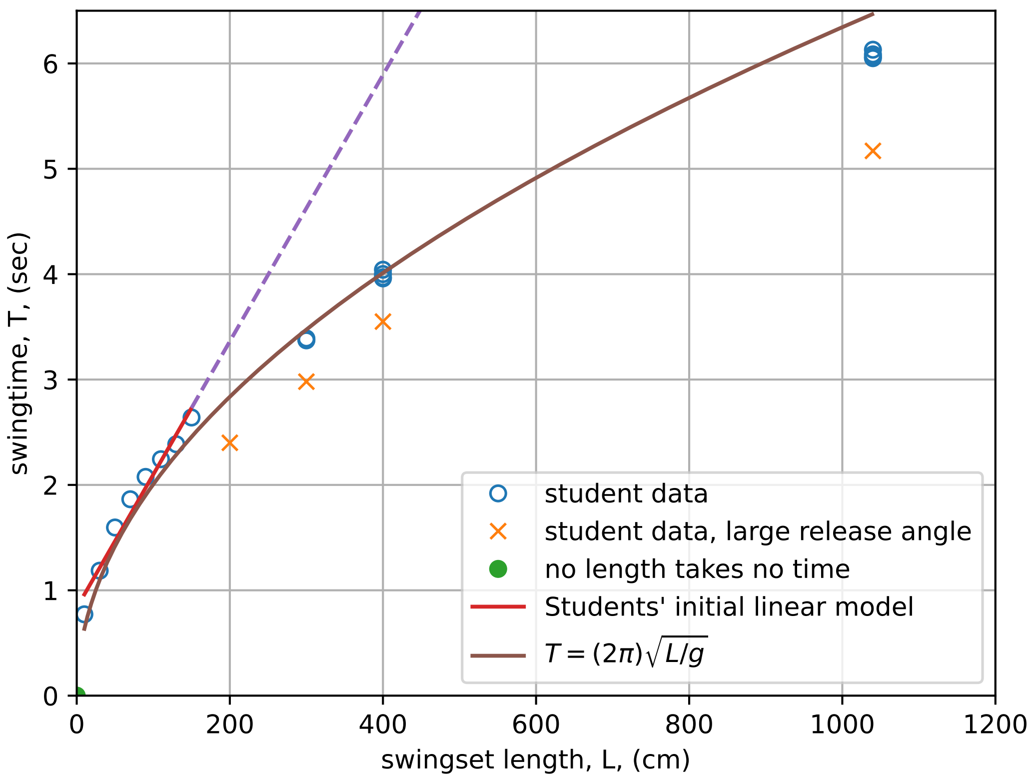

Figure 2. Typical student data, using metal washers and string to model the motion of a swingset. Students are coached to measure the time associated with multiple "there and back" swings and then divide to find an average time so that their reaction times (typically $\approx 0.3 seconds$ via [6] can be ignored.

measure as a 3-story high swingset. As seen in figure 3, even this data is reasonably described by the linearized model. Note, most of this work depends on the students using a relatively small initial release angle (unlike the Kiiking example). If students decide to release a swingset from horizontal, the $\sqrt{L}$ dependence will be harder to see, which is again visible in figure 3.

2. Predicting the height of a swingset

With a reliable model for swingsets developed, a common follow-up assignment is to estimate the height of a bridge or arch from video of people swinging below it. Two specific examples we've used, one in Sault Ste. Marie, Canada [7], and another near Moab, Utah [8], are available on youtube.

At this point, from their data analysis, students know that swingtime, T, and swingset length, L, are described by an expression similar to the following (with slightly

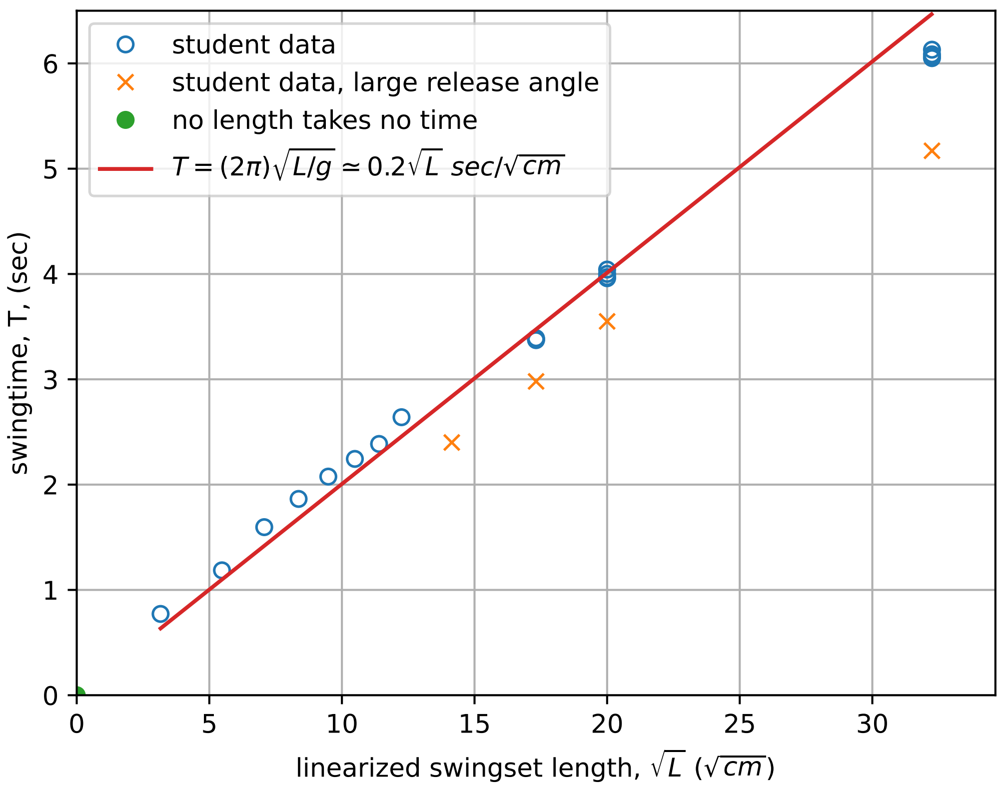

Figure 3. Linearized swingset data from a class. Note that $200cm$, $300cm$ and $400cm$ lengths are included and described well by the $\sqrt{L}$ trendline – so long as the initial release angle is small. The "almost no length" swingset that takes "almost no time" to swing is also nicely included.

different slopes, depending on student measurements):

$$T \simeq \left(0.20 \frac{seconds}{\sqrt{cm}}\right) \sqrt{L}, or \tag{1}$$

$$L \simeq \left(25 \frac{cm}{sec^2}\right) T^2. \tag{2}$$

Then, if a student is able to measure the time corresponding to half or a quarter of a swing from the video, they can use the model to predict the swing's height. As an example, the man in the Sault Ste. Marie video seems to take a little less than 4 seconds to make half a swing (from 0:36 to 0:40 in the video). With a period of $T \approx 7 \pm 1s$, students can predict a swingset length of about $L \approx [900 - 1600]cm \approx 12 \pm 4m$. A more detailed calculation using video analysis [9] gives a swingset length of $11.35 \pm 0.3m$. The video in Moab can be similarly analyzed, although estimating even a quarter period in the Moab video is challenging..

3. Evaluating answer using Open Source Intelligence

Over the past few years, news organizations using "open source intelligence techniques" have gained some notoriety. One such organization, "Bellingcat" [10], among many investigations, gained prominence by revealing a plausible method by which Russian opposition political Alexey Navalny was poisoned in 2020 [11]. This organization uses "open source" information to engage in investigative journalism and fact-checking. On their website, Bellingcat makes a large number of informational "howto" articles available [12] .

Along similar lines, an unclassified 1996 copy of the United States Government Defense Mapping Agency's "Student Handbook on Photo Interpretation Principles" is available in many places online [13]. This book, written for novice cartographers, illustrates many ways to infer measurements based on aerial and satellite photographs. This is a useful reference, given the abundance of information that Google Maps and similar sites make available.

Both of these resources investigate and test claims using information that is freely available to anyone with an internet connection. In the Physics classroom, this seems like an exciting, modern, and relevant place to make a connection to "checking our answer" with techniques similar to those use to establish that chemical weapons were being used in the Syrian Civil war [14].

So at the end of this lab, we work to use some other way of knowing to check our answer for the height of the Moab arch or the pedestrian bridge in Sault Ste. Marie. Here are a few of the conversations and student efforts that have resulted:

3.1. screen grab and scaling

Take a screenshot from the video, assume person is 5'8" ??? and then scale pixels to estimate height of bridge/arch

Or pine tree?

3.2. geographic searching

Many students have done a google search for "height of Sault Ste. Marie bridge" or similar. Google seems to interpret this as the height of the "International" road bridge for vehicles across the Soo locks [15], which is reported to be 124 feet (38m) in height. This is obviously not the correct bridge in the video.

Along similar lines, I'm embarrassed to admit that some of my students have asked the video creator the bridge's height.

3.3. shadows

There are many shadows cast in the overhead photos of Sault Ste. Marie. We can use some to estimate, by scaling, the height of the bridge to be:

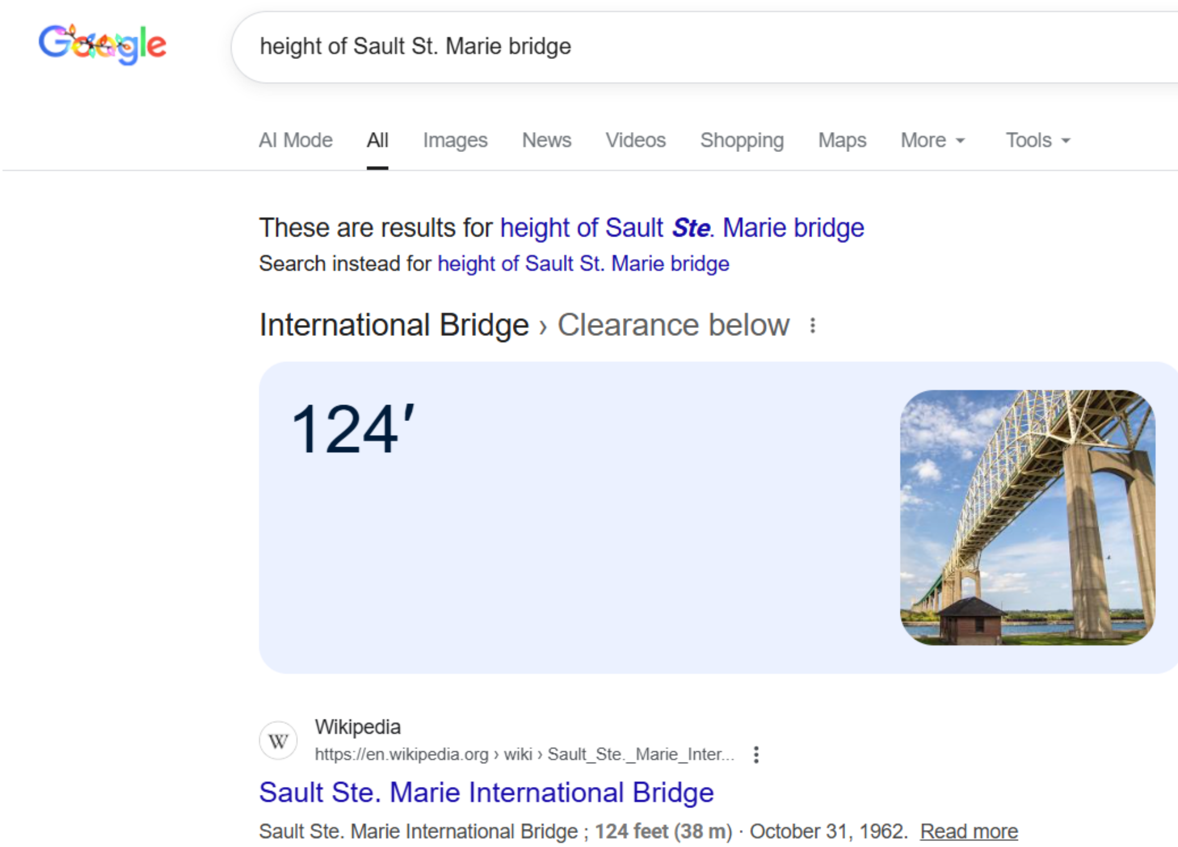

Figure 4. Google results for "height of Sault Ste. Marie bridge" are obviously incorrect for the pedestrian/bicycle bridge in question.

3.4. geographic clues

The Moab arch video was taken at Arches National park. Some students claim that the video matches the Gateway Arch??? which has known height.

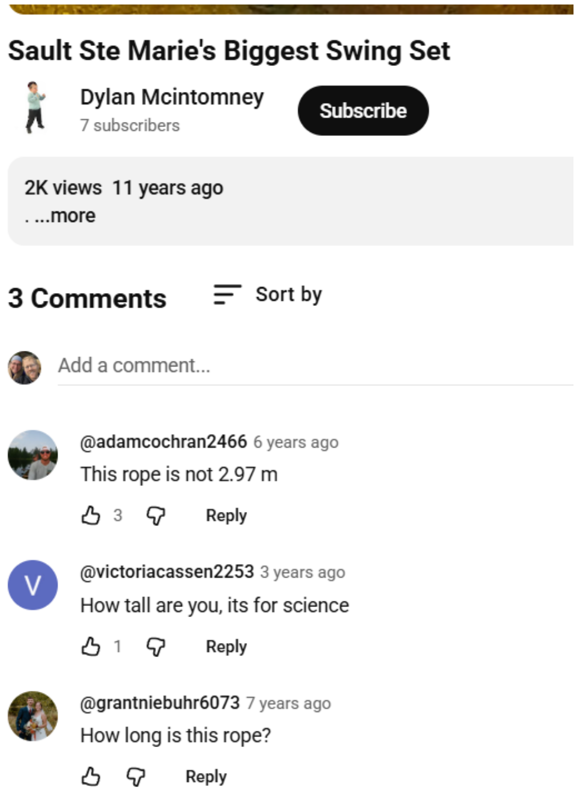

Figure 5. Youtube video comments, from my students, on https://www.youtube.com/watch?v=arxiDyDouHo

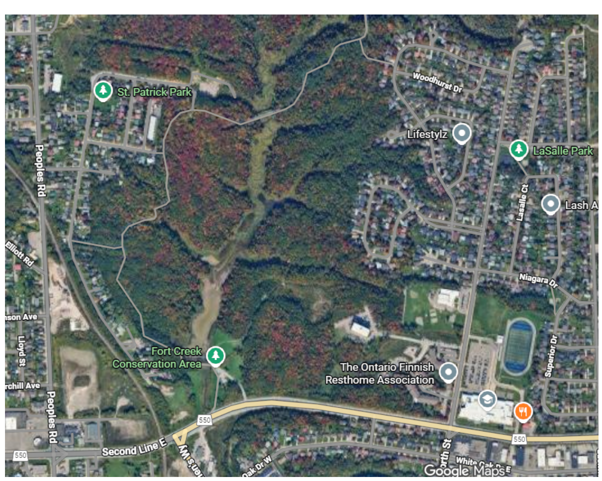

Figure 6. Sault Ste. Marie, Ontario, Canada. 46.543499, -84.341888 The Hub trail bridge and football field are about 1000meters apart.

Figure 7. We are fairly certain this is the pedestrian bridge in the video [7]. Hub Trail Sault Ste. Marie, ON P6C 3A5, Canada 46.546738, -84.342165

Figure 8. Using Google Maps to measure the length of the Bridge deck's shadow.

Figure 9. The shadow cast by the field goalposts on an (assumed) American football field, located about 1km from the Sault Ste. Marie pedestrian bridge in this video [7] 750 North St Sault Ste. Marie, ON P6B 2C5, Canada 46.541034, -84.332469

3.5. Conclusion

Table **??** gives a summary of swingset/bridge/arch heights from various open source measurement approaches. What's the real height of the Hub Trail Bridge in Sault Ste. Marie? The answer varies: Is there snow on the ground, is the creek up because of flooding, or has the grass been beaten down by the wind?

length being measured/calculated	**method**	**measurement**
Swingset rope, equation 2	Youtube time	$12 \pm 4m$
Swingset rope, equation 2	Video Analysis time	$11.35 \pm 0.3m$
Bridge support tower	pixel scaling, man as reference	???
Bridge deck above valley	shadows, football field?	???
Bridge deck above valley	shadows, some other calibration???	???
Valley depth	topographic map	

Table 1. Bridge height estimates

References

[1] College Physics 2e 16.4 The Simple Pendulum https://openstax.org/books/college-physics-2e/pages/16-4-the-simple-pendulum

[2] Pivot Interactives Library: Pendulum Investigation 2025 Pivot Interactives, SBC Version 1.252.0 https://www.pivotinteractives.com/

[3] https://www.youtube.com/watch?v=UtRRHrEaBBA Kiiking. Meanwhile in Finland

[4] https://commons.wikimedia.org/wiki/File:Fragonard,_The_Swing.jpg Jean-Honoré Fragonard: The Swing 1767

[5] https://en.wikipedia.org/wiki/Swing_(seat) Swing (seat)

[6] https://humanbenchmark.com/tests/reactiontime

[7] https://www.youtube.com/watch?v=arxiDyDouHo Sault Ste Marie's Biggest Swing Set Dylan Mcintomney Oct 29, 2014

[8] https://www.youtube.com/watch?v=4B36Lr0Unp4 World's Largest Rope Swing devinsupertramp (Devin Graham) Feb 15, 2012

[9] If you download the video from YouTube and import it into Vernier's LoggerPro, it looks like a half swing runs from 36.603s to 39.973s, a $T/2 = 3.37s$ or $T \approx 6.74s$. Video frames are 0.034s apart so it might be reasonable to assign an uncertainty of 0.07seconds to this measurement. Using the small angle approximation in equation 2 then gives an estimate swingset length of $L \approx \left(25\frac{cm}{sec^2}\right)(6.74 \pm 0.07s)^2 = [1112, 1159]cm \approx 11.35m \pm 0.3m$

[10] https://www.bellingcat.com/ and https://en.wikipedia.org/wiki/Bellingcat

[11] https://www.bellingcat.com/resources/2020/12/14/navalny-fsb-methodology/.

[12] https://www.bellingcat.com/category/resources/

[13] Photo Interpretation Student Handbook. United States Defense Mapping Agency, Unclassified, March 1996.

[14] "Brown Moses" was a precursor to the Bellingcat website, run by the same group of people. https://brown-moses.blogspot.com/2014/04/evidence-from-2-weeks-of-chlorine.html and https://en.wikipedia.org/wiki/Bellingcat see footnote 21

[15] https://en.wikipedia.org/wiki/Sault_Ste._Marie_International_Bridge